

Classification using ANN and CNN of Hot Dog vs. Not Hot Dog

M RAYAN UR REHMAN KHAN
Faculty of Computer Science and Engineering
Ghulam Ishaq Khan Institute (GIKI)
Topi, Pakistan
Registration No: 2023281

Abstract—This report details the development of an image classification system designed to distinguish between hot dogs and other food items. We compare a basic Artificial Neural Network (ANN) with a Convolutional Neural Network (CNN). Our findings demonstrate that while ANNs struggle with spatial hierarchies, CNNs using Dropout and Data Augmentation provide significantly higher validation accuracy.

Index Terms—Deep Learning, CNN, Overfitting, TensorFlow, Image Classification.

I. INTRODUCTION

The "Hot Dog / Not Hot Dog" problem is a binary classification task in computer vision. Using TensorFlow version 2.20.0, this project implements and optimizes deep learning models to achieve reliable real-world performance.

II. DATA EXPLORATION AND VISUALIZATION

The dataset consists of two classes where label 0 represents "hot_dog" and label 1 represents "not_hot_dog".

- **Balance:** The training set contains 249 images per class, and the test set contains 250 images per class.
- **Image Properties:** Images were resized to $64 \times 64 \times 3$ (RGB) for computational efficiency.



Fig. 1. Sample images from the Training Dataset.

III. ANN MODEL ANALYSIS

A simple ANN was constructed by flattening the input pixels into a 1D vector followed by Dense layers.

- **Performance:** The ANN achieved a final validation accuracy of 57.14% after 10 epochs.

- **Failure Analysis:** ANNs perform poorly on image data because flattening destroys the spatial relationship between neighboring pixels.

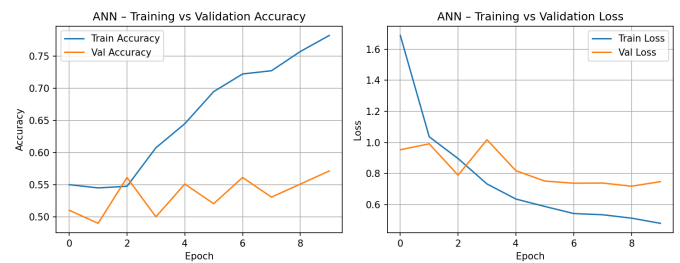


Fig. 2. ANN Training vs. Validation Accuracy and Loss.

IV. CNN IMPLEMENTATION AND COMPARISON

A CNN was implemented using three *Conv2D* layers and *MaxPooling2D*.

- **Performance:** The base CNN reached a final validation accuracy of 58.16% within 15 epochs.
- **Advantage:** CNNs use kernels to extract local features, making them more robust for visual recognition.

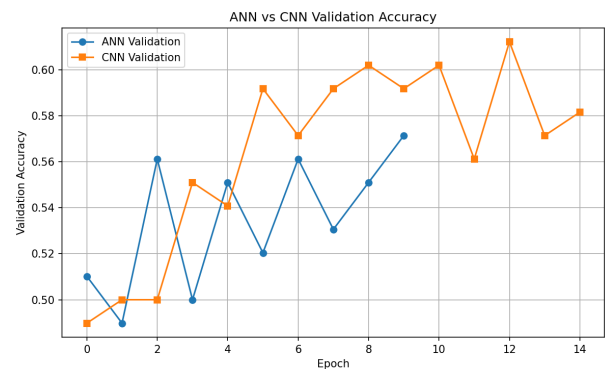


Fig. 3. Validation Accuracy Comparison: ANN vs. CNN.

V. FIXING OVERFITTING

To address overfitting, we introduced two primary fixes:

- 1) **Dropout:** Randomly deactivating neurons during training to prevent co-adaptation.

2) **Data Augmentation:** Applying rotations and flips to simulate a larger dataset.

These techniques improved the final validation accuracy to 67.35%.

VII. CONCLUSION

This project demonstrated the transition from simple ANNs to optimized CNNs. By addressing overfitting, we developed a model capable of distinguishing hot dogs in real-world scenarios.

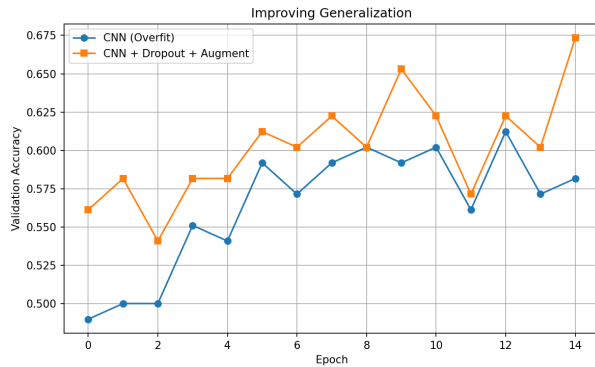


Fig. 4. Generalization improvement using Dropout and Augmentation.

VI. REAL-WORLD TESTING AND YOLO EXTENSION

A. Single Image Prediction

The final model correctly identified a real-world test image as a **HOT DOG** with a confidence score of 0.60.



Fig. 5. Model testing on a real-world image.

B. YOLO Object Detection

For the extra credit task, a pre-trained *YOLOv8n* model successfully detected the "hot dog" class with 0.76 confidence.

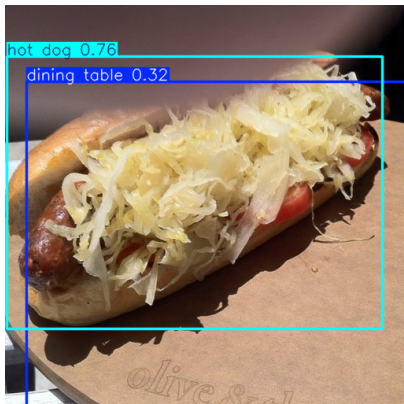


Fig. 6. YOLOv8 Detection Result.